

Enhancing Drug Repurposing Accuracy through AI-Orchestrated Integration of Knowledge Graph Reasoning, Structure-base Model Prediction, and Gene Expression Analysis



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Introduction

Drug repurposing enables faster and lower-risk drug development. However, current AI approaches remain **fragmented**, lacking a unified framework for **multi-source evidence integration**.

To address this, we present an **AI-orchestrated platform** that integrates **three evidence layers**: **structure-based modeling** of drug–target interactions, **knowledge graph reasoning** for disease relevance, and **gene-expression analysis** for functional validation. Each component generates independent evidence and is integrated at the **decision level** for **cross-validation across heterogeneous data**.

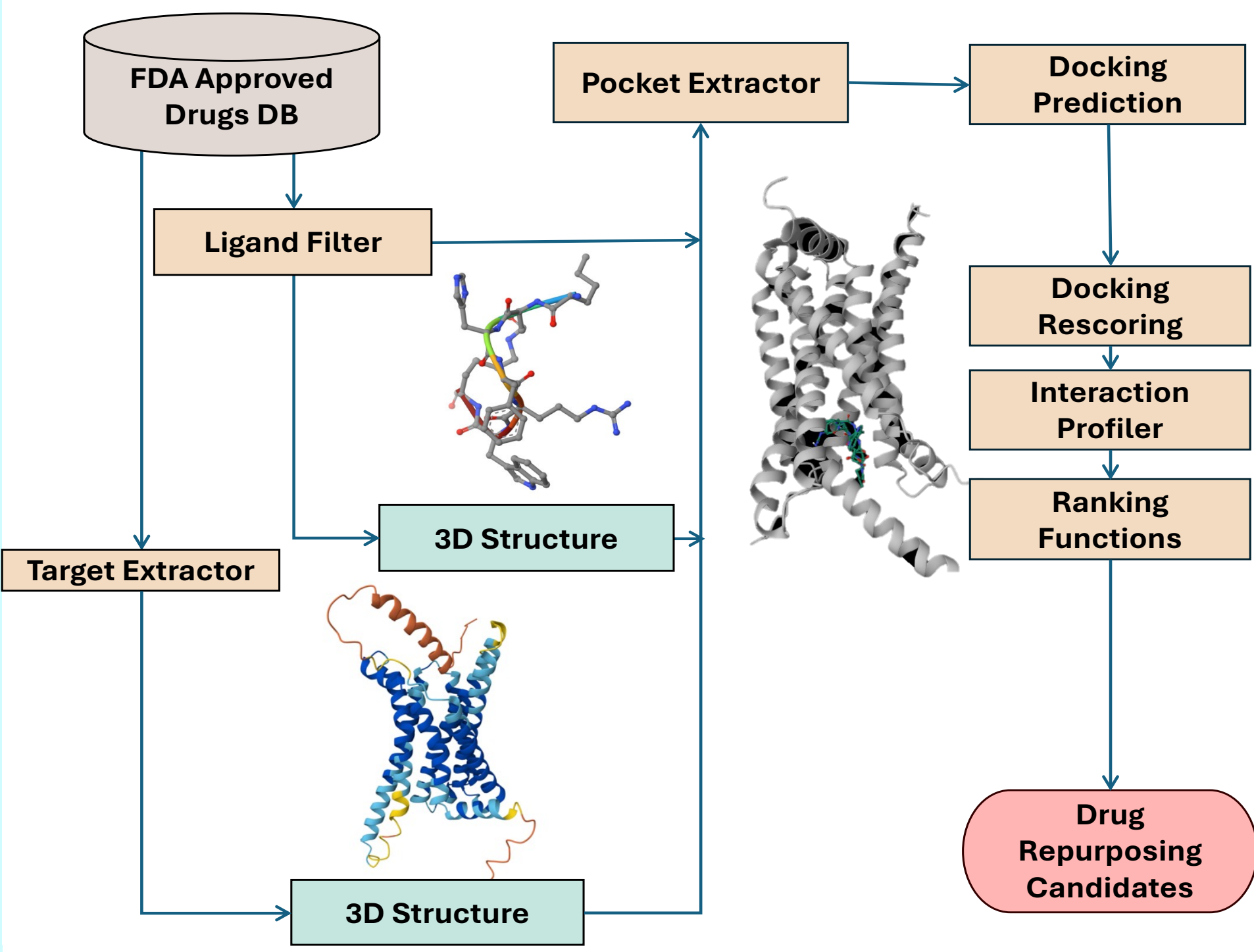
The platform supports **two workflows**: a **Drug–Target Driven** workflow that starts from existing drugs to identify new targets and indications, and a **Disease–Gene Driven** workflow that begins from disease biology to discover candidate therapeutics.

By combining **multi-modal evidence integration with bidirectional exploration**, the framework improves **robustness, interpretability, and translational relevance**, transforming fragmented AI predictions into **coherent and actionable therapeutic insights**.

Multi-Modal Evidence Platform

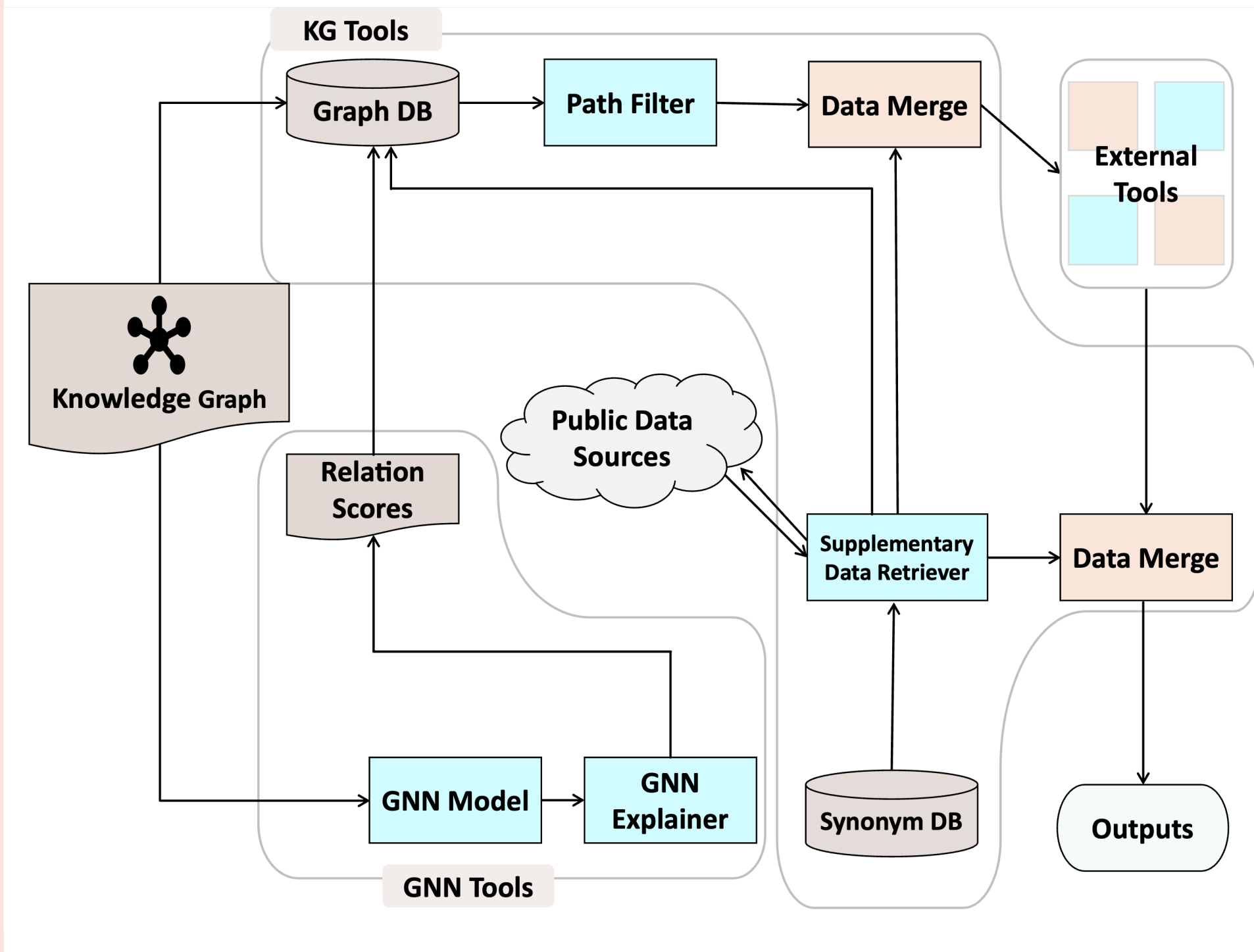
Structure-based Modeling

Structure-based prediction of drug–target interactions



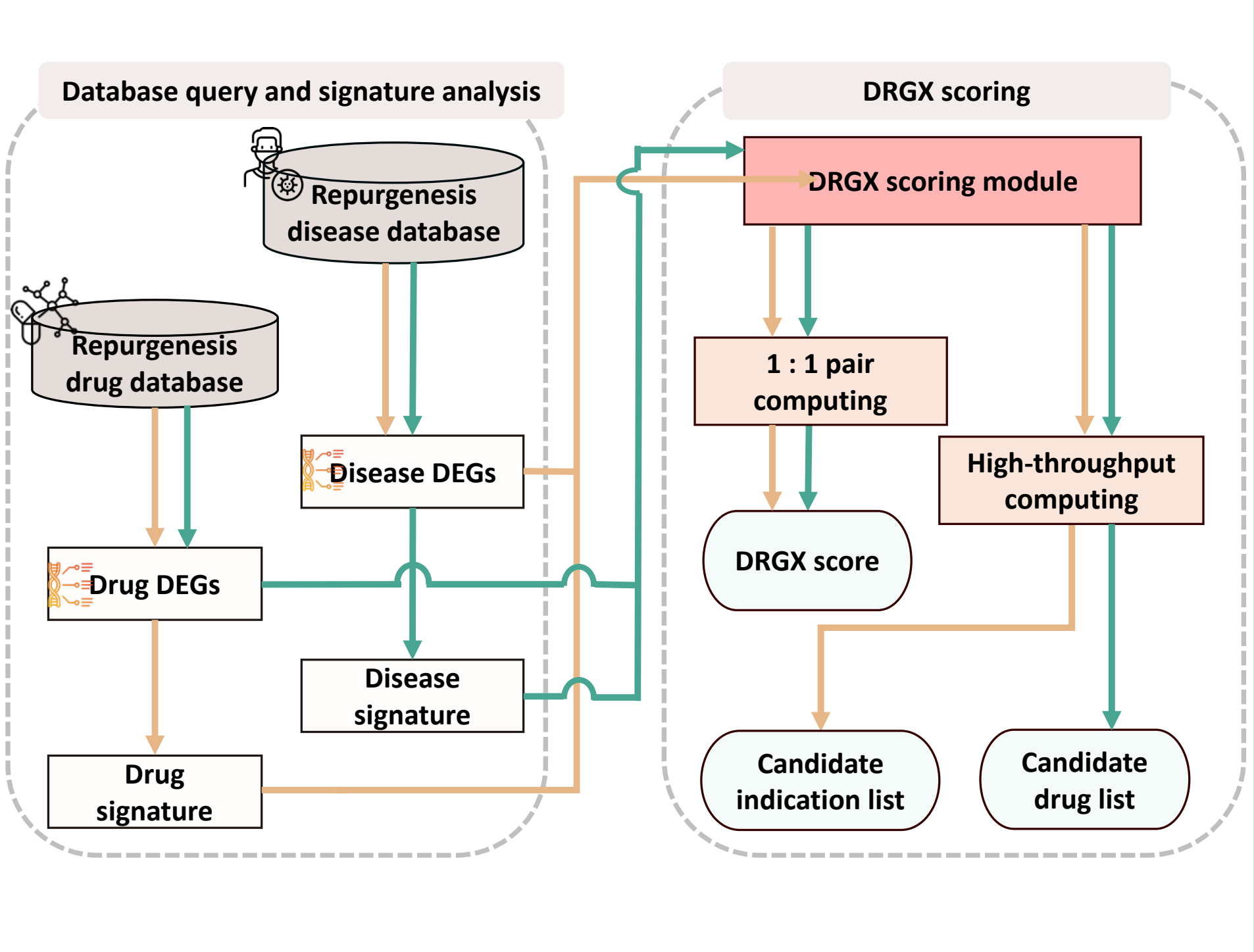
Knowledge Graph Reasoning

Knowledge graph reasoning for disease relevance



Gene Expression Analysis

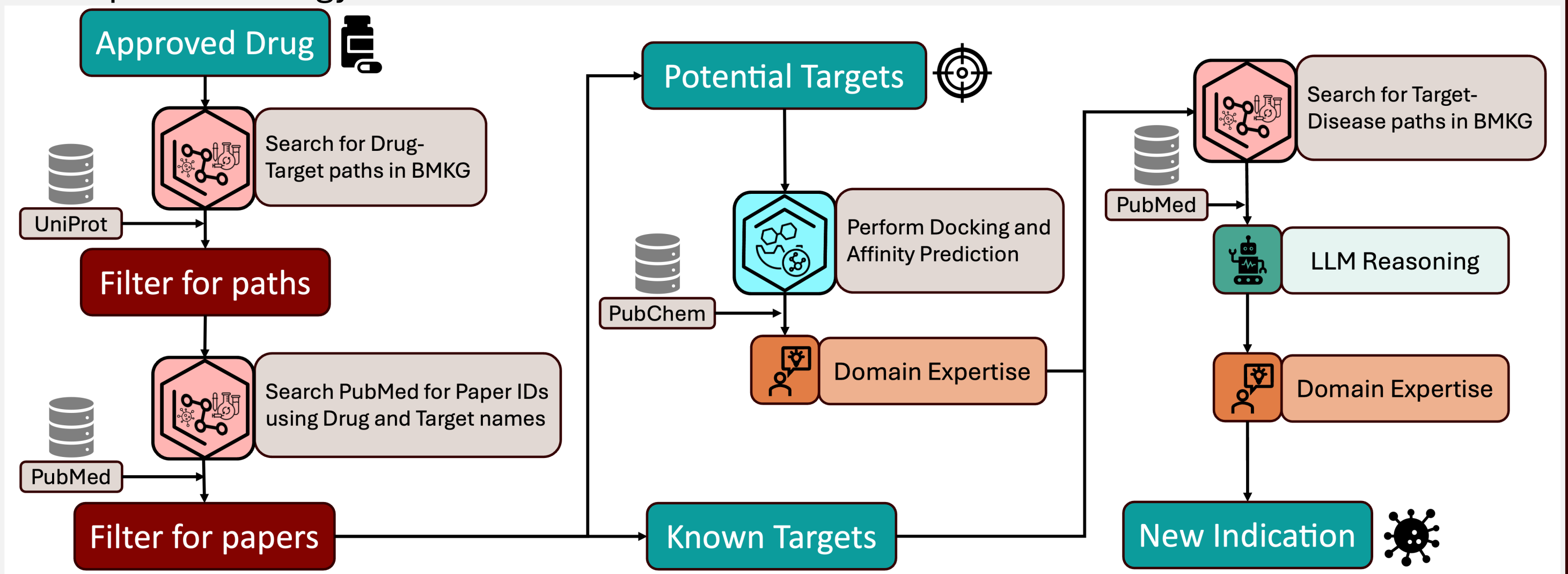
Gene-expression signature matching



Orchestrated Intelligence Workflow

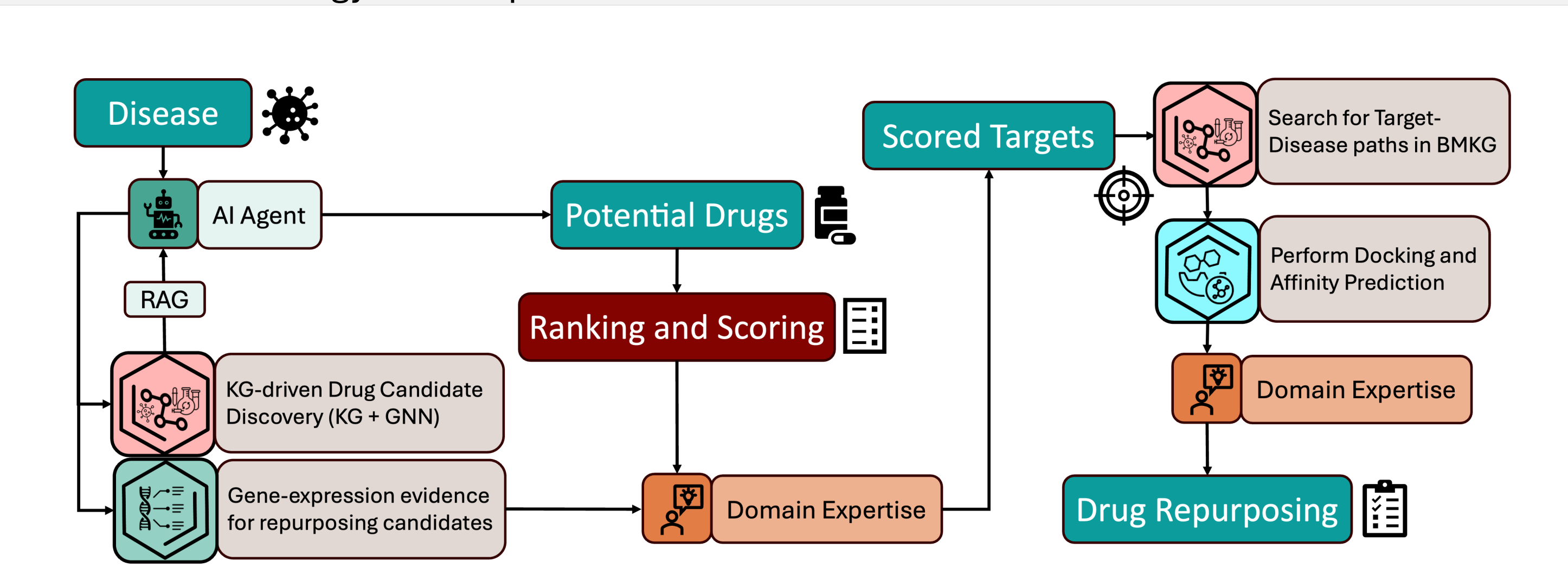
Drug–Target Driven Discovery

From pharmacology to indication



Disease–Gene Driven Discovery

From disease biology to therapeutics



Translational Pipeline: From Discovery to IP

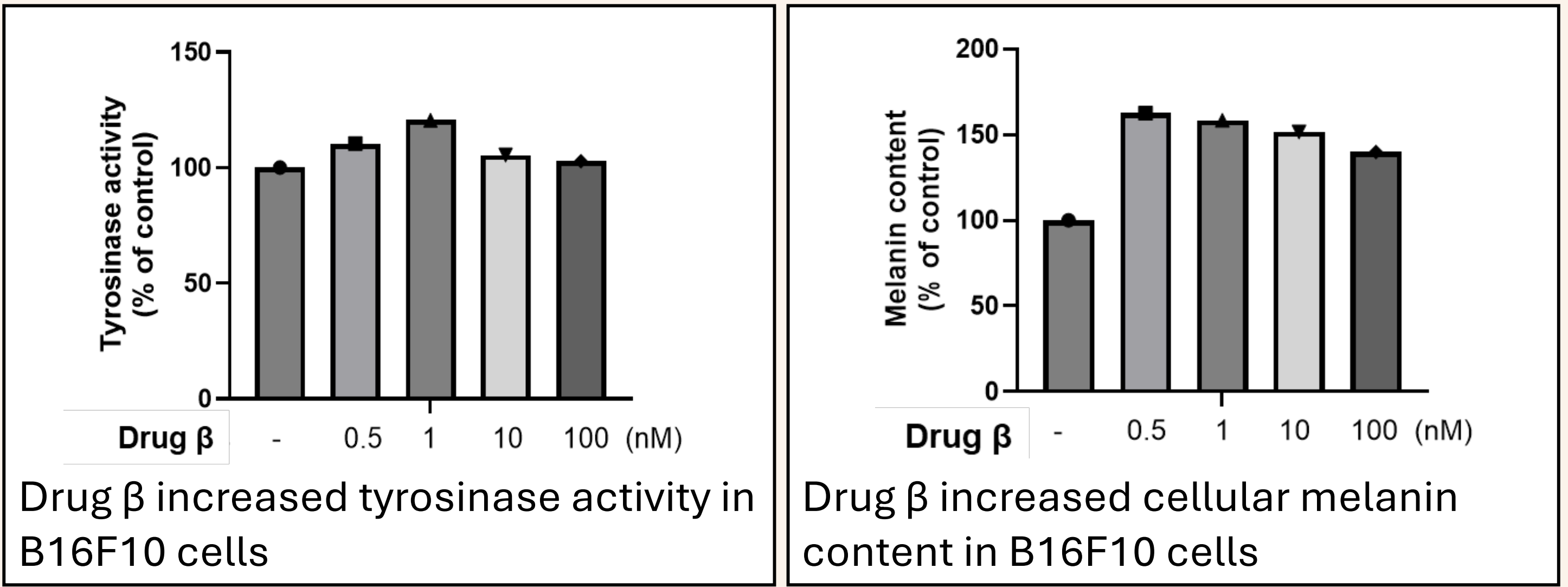
AI-identified candidates advancing across translational stages

Disease Area	Indication	Drug ID	PoC Validation	IP Protection	Pre-Clinical
Autoimmune	Vitiligo				
Fibrosis	Idiopathic Pulmonary Fibrosis				
Neurodegenerative	Spinocerebellar Ataxia (Rare)				

Leveraging existing safety profiles, our pipeline accelerates repurposing from AI discovery to clinical and IP outcomes.

Case Study: Drug β Repurposing

Drug β Repurposing for Vitiligo Treatment (Patented)



We are thankful to R&D and DevOps team for their continued support in this work

